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TRANSPORT PHENOMENA II, CHE 312

QUIZ 2

CLOSED BOOK, TIME: 20 MINUTES

A cold liquid film is flowing down a vertical solid wall with the velocity profile

$$v_z = v_{z,max} \left[1 - \left(\frac{x}{\delta} \right)^2 \right]$$



The liquid film, initially at a uniform temperature T_1 comes into contact with a warm gas that maintains the gas-liquid interface temperature at T_0 .

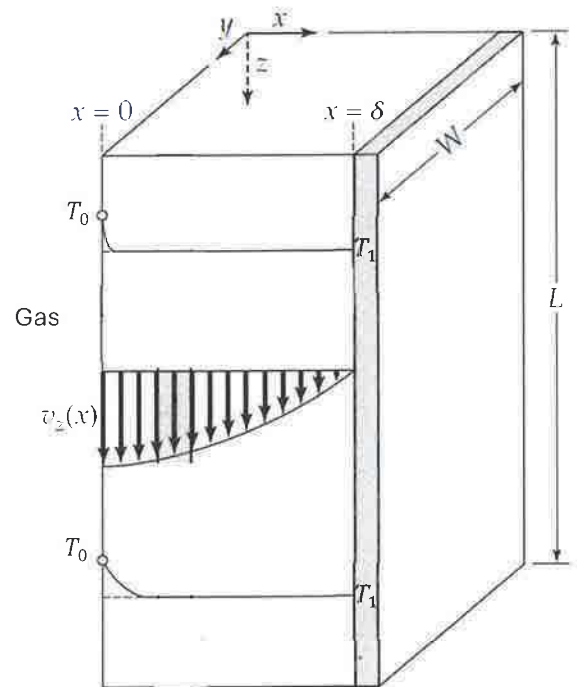
Postulates:

- Liquid has constant density ρ , heat capacity c_p , and thermal conductivity k .
- Short contact times, so the liquid temperature changes appreciably only in the immediate vicinity of the gas interface.
- For $\frac{x}{\delta} \ll 1$ assume $v_z \approx v_{z,max}$

- 1- Simplify the equation of energy to derive a partial differential equation (PDE) for the steady-state dimensionless temperature $\theta = \frac{T - T_1}{T_0 - T_1}$ as a function of x and z .

- 2- Convert the above PDE to an ordinary differential equation for θ as a function of $\eta = \frac{x}{\sqrt{4\beta z}}$, where $\beta = \frac{k}{\rho c_p v_{z,max}}$.

- 3- Write down two boundary conditions for $\theta = \theta(\eta)$.



*θ is a function of η
and η is a function
of x and z .*

§B.9 THE EQUATION OF ENERGY FOR PURE NEWTONIAN FLUIDS WITH CONSTANT^a ρ AND k

$$[\rho \hat{C}_p DT/Dt = k \nabla^2 T + \mu \Phi_v]$$

Cartesian coordinates (x, y, z) :

$$\rho \hat{C}_p \left(\frac{\partial T}{\partial t} + v_x \frac{\partial T}{\partial x} + v_y \frac{\partial T}{\partial y} + v_z \frac{\partial T}{\partial z} \right) = k \left[\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} \right] + \mu \Phi_v \quad (\text{B.9-1})^b$$

Name:

velocity is a function of z $v_z(x)$
 $v_y = v_x = 0$

Temp: $T(z, x)$

Boundary conditions: $T(z \rightarrow \infty, x) = T_1$
 $T(z, x=0) = T_0$
 $T(z, x \rightarrow \infty) = T_0$

negligible

$$\rho c_p \left[\frac{\partial T}{\partial t} + v_x \frac{\partial T}{\partial x} + v_y \frac{\partial T}{\partial y} + v_z \frac{\partial T}{\partial z} \right] = k \left[\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} \right] + \mu \phi$$

@ S.S. $v_x = 0$ $v_y = 0$ $v_z = 0$ $\frac{\partial^2 T}{\partial x^2} = 0$ $\frac{\partial^2 T}{\partial y^2} = 0$ $\frac{\partial^2 T}{\partial z^2}$ negligible $\mu \phi$

$$\rho c_p v_z \frac{\partial T}{\partial z} = k \frac{\partial^2 T}{\partial y^2}$$

1) good!
 3) +10.

2) -20

$v_z(x)$ we can rewrite in terms of

$$v_z = \left(\frac{\rho g \delta}{\mu} \right) x$$

$$\rho c_p \frac{\rho g \delta}{\mu} x \frac{\partial T}{\partial z} - k \frac{\partial^2 T}{\partial y^2} = 0$$

$$x \frac{\partial T}{\partial z} = \frac{k \mu}{\rho^2 c_p g \delta} \frac{\partial^2 T}{\partial y^2}$$

β

$$\frac{y}{\delta} \ll 1$$

$$x \frac{\partial T}{\partial z} = \beta \frac{\partial^2 T}{\partial y^2}$$